

INSIDE SET-TOP BOX

STBs have enhanced our TV viewing experience through a host of value-added services. This article takes you deep inside a set-top box to provide you a complete overview of its architecture

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With the advent of digital television, set-top box (STB) has become a buzzword. While transmission of television signals has gone digital, most TVs are still analogue. Therefore an STB is required to convert the digital signal to analogue signal so as to enable a viewer to watch digital content on the TV set. However, an STB is not merely a digital to analogue signal converter, it also provides a host of other services to enhance the TV viewing experience—provision of electronic programme guide to keep track of the content is one such example.

This article aims to take you deep into the set-top box to enhance your understanding about the STB hardware architecture.

STB hardware components

The STB mainly consists of following components:

Antenna. The antenna is not essentially the part of a set-top box, as it is fitted outside an STB. It contains low-noise block (LNB) that receive the satellite signal with a frequency of approximately 12 GHz and performs a down conversion to a frequency range of 950 to 1450 MHz. Subsequently, the signal is passed on to the 'front end' inside the set-top box.

Power supply. Power supply provides the required voltage to run various peripherals inside an STB. In some cases, it can be used to power mechanised antennae to align them in various directions in order to receive different signals.

Front panel. It is a small micro-controller-based hardware peripheral inside an STB that receives the user

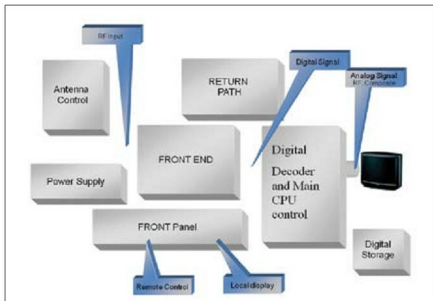


Fig. 1: Various hardware components inside an STB

input and interacts directly with the main CPU control (processor). It contains an IR receiver to listen to remote control input. Also, it is directly linked to some input buttons or keys present on the outer frame of an STB.

Front end. It is also known as Network Interface Module (NIM). Front end is a peripheral device that interacts with the antenna to receive the down converted signal and send it to digital decoder in the form of a transport stream. Basically, the front end consists of two major components—tuner and demodulator. The tuner listens to the signal received from the antenna and passes it on to the demodulator for generating the transport stream. Different types of NIMs are satellite QPSK, cable QAM, terrestrial COFDM, 8-VSB and telco ADSL.

Digital decoder. A digital decoder receives the transport stream sent by the demodulator inside the front end and then demultiplexes and decompresses

the requested video by the user. It then converts the digital stream to an analogue signal, which is sent to RF port. The task of the demultiplexer or a demux is to separate all the data streams packaged together by the HeadEnd in the specified format. The data contains audio, video and DVB (digital video broadcasting) specified data like SI/PSI tables. After the demuxing, decompression is done by a specified decoder in specified formats like MPEG-2 or MPEG-4. The specific data or information is given to CPU for processing for advanced features and memory storage and to be used in graphics.

CPU (Processor). Just as in a regular PC, the processor inside the CPU takes care of interactions of all the hardware peripherals and software modules inside a set-top box. It manages the Real-Time Operating System (RTOS). The processor also takes care of memory devices and other forms of digital stor-